SHEMILKUMAR E A

Frontend Developer

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PROFESSIONAL SUMMARY

Frontend Developer with 5 years of professional experience building scalable, high-performance web applications using **Angular and React**. Specialized in real-time data visualization, WebSocket-based systems, component-driven architecture, and performance optimization. Delivered production-grade platforms in medical domains handling high-frequency ECG data streams and multi-source clinical datasets. Experienced in TypeScript, RxJS, SCSS, Tailwind CSS, REST API integration, and Agile/Scrum delivery. Strong debugger with a track record of resolving complex UI and data synchronization issues in production systems.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS

Frameworks & Libraries: Angular (v16–v20), React.js, Next.js, RxJS, Angular Material, Tailwind CSS

Real-Time & Data: WebSocket, High-Frequency Data Streaming, Buffering, Data Synchronization, Interactive Charting

Architecture: Component-Driven Development, Modular Architecture, Reusable Component Libraries, API Service Layer, Scalable Folder Structure

Performance: Lazy Loading, Code Splitting, Tree Shaking, Bundle Optimization, Change Detection Strategy, OnPush, Virtual Scrolling

Tools & Platforms: Git, GitHub, Figma, Vite, REST APIs, Postman, VS Code

Process: Agile, Scrum, Sprint Planning, Code Reviews, Technical Mentoring

WORK EXPERIENCE

Frontend Developer | Probeplus Innovative Solutions Pvt. Ltd. | Bengaluru, India*Jun 2023 – Present*

- Architected and built a **real-time ECG monitoring platform** using Angular and WebSocket — streaming continuous high-frequency biosensor data with custom buffering and automatic reconnection to handle network interruptions without data loss.
- Built a **clinical data analysis and annotation platform** (MIMIC dataset) for doctors to analyze time-synchronized multi-source patient data including ECG signals, vital signs, and clinical event logs.
- Designed scalable, **modular Angular architecture** from scratch — defined folder structure, module boundaries, component hierarchy, and shared service patterns, reducing code duplication by **~40%**.
- Built a centralized API service layer standardizing HTTP handling, error states, loading states, and response mapping — reducing redundant API code by **~50%**.
- Developed a library of 30+ reusable Angular components with clearly defined @Input/@Output contracts, reducing development time for new features significantly.
- Executed Angular version migrations from v12 to v16 to v20 — updated deprecated APIs, migrated RxJS operators, and ensured zero regression across the full application.
- Optimized application performance: reduced initial load time by **35%** and bundle size by **20%** through lazy loading, code splitting, OnPush change detection, and bundle analysis.
- Coordinated frontend delivery across a team of five developers — led sprint planning, task breakdown, and code reviews to improve delivery consistency.
- Mentored junior developers on Angular best practices, TypeScript usage, and coding standards, reducing production bugs by ~20%.

Frontend Developer | Tata Consultancy Services | New Delhi, India*May 2021 – Jun 2023*

- Built and maintained 15+ responsive, cross-browser-compatible UI components using React.js, JavaScript, and modern CSS across multiple client-facing projects.
- Integrated 10+ REST APIs with complete handling of data transformation, error states, loading states, and edge cases.
- Implemented modular React component structures and shared utility patterns, reducing code duplication by ~20% across modules.
- Improved page rendering performance by ~15% through optimized component lifecycle, memoization, and efficient asset handling.
- Contributed to Agile/Scrum ceremonies — sprint planning, daily standups, code reviews — and consistently delivered features on schedule.

KEY PROJECTS

Real-Time ECG Monitoring & Visualization Platform — *Angular • WebSocket • RxJS • SCSS • TypeScript*

- Designed and built the full frontend from scratch — architecture, data flow, component structure, and WebSocket integration with no existing codebase to reference.
- Implemented WebSocket-based real-time data streaming with a custom buffering layer to handle temporary network drops and resync missed ECG packets seamlessly on reconnection.
- Rendered multiple simultaneous ECG leads (ECG I, II, V1, etc.) alongside live vitals — Heart Rate, SpO2, RR, IBP, Pulse Rate — with synchronized real-time updates across all views.
- Optimized high-frequency rendering pipeline to sustain smooth graph animation under continuous data load without UI lag or frame drops.
- Built a patient list panel with live per-patient vitals, alert states (e.g. NIBP HIGH), and bed/unit assignment — supporting multi-patient monitoring in ICU/TICU contexts.
- Implemented timeline-based navigation enabling doctors to select and inspect ECG data at precise timestamps for clinical decision-making.

Clinical Data Analysis & Annotation Platform — *Angular • RxJS • TypeScript • Tailwind CSS*

- Built a doctor-facing platform to analyze multi-source patient data from the MIMIC clinical dataset — ECG signals, vital metrics, and clinical event logs — in a unified time-synchronized interface.
- Implemented an interactive clinical event timeline covering admissions, procedures, prescriptions, and chart events with time-block navigation across large medical datasets.
- Built dynamic charts for vitals (HR, SpO2, RR) and ECG with zoom, time-range selection, and macro/micro level inspection for detailed analysis.
- Designed an annotation pipeline enabling doctors to assign criticality scores and structured clinical reasoning — outputs formatted as labeled datasets for AI model training.

Scalable Enterprise Angular Platform — *Angular • SCSS • TypeScript • RxJS*

- Architected a scalable Angular platform with reusable component library (30+ components), shared services, and clearly defined module boundaries across multiple feature teams.
- Established coding standards, API service patterns, and folder structure conventions adopted across the full engineering team.

EDUCATION

Master of Computer Applications — Computer Science *2023*

Jain University, Bengaluru, India